GEP: A Post-Quantum Cryptosystem from Exceptional Algebra

Key Exchange and Encryption via the Albert Algebra $J_3(\mathbb{O})$
and the Non-Associative Hidden Subgroup Problem for F_4

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Formal verification: Lean 4, Coq, Agda, Isabelle/HOL, Rust. Repository: `github.com/SNAPKITTYWEST/gep-formal` (private).

Abstract

We present **GEP** (Galois Encryption Protocol), a novel post-quantum cryptographic primitive whose security rests on a problem for which no efficient classical *or* quantum algorithm is known: the **Non-Associative Hidden Subgroup Problem (NA-HSP)** for the exceptional Lie group $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$.

Plaintext is embedded as a diagonal element of the 27-dimensional Albert algebra $J_3(\mathbb{O})$. Encryption applies a secret sequence of 16 F_4 rotations—the *collapse operator* φ—spinning the matrix into a high-entropy off-diagonal state we call *Nūn-space*. Decryption applies φ to recover the diagonal. The security of φ-inversion reduces to NA-HSP for F_4: a problem the quantum Fourier transform cannot efficiently attack because the octonions are *non-associative* and admit no faithful finite-dimensional linear representation.

We contrast GEP with the **Fibonacci Braid Conjugacy (FBC)** key exchange, which we break in polynomial time via a Sylvester equation, demonstrating precisely why representation-theoretic keys fail. GEP avoids this attack by construction: the shared secret is derived from the full non-associative structure, not from any matrix representation.

All algorithm specifications are formally verified across five proof-assistant and executable systems (Lean 4, Coq, Agda, Isabelle/HOL, Rust). Performance targets: φ evaluation in $16.7\,\mu$s scalar, $2.7\,\mu$s AVX-512 on commodity hardware. Post-quantum security level: NIST Level 5+ (classical 2^{2048}, quantum 2^{1042} query×gate complexity).

This is a research prototype. The conjectures in Section 6.5 are unproven. Do not deploy in production.

Contents

1 Author's Introduction

[Ahmad's opening — approximately 3 pages. Ahmad writes this section in his own voice. The mathematical sections that follow handle all proofs, specifications, and verification results.]

Suggested arc: (1) Why you built GEP on a phone with Ollama and no frontier models — the origin of the formal verification instinct. (2) The octonion insight: the structure that stops Shor cold. (3) What Nūn-space means beyond mathematics — the cosmology behind the name. (4) The transition sentence handing off to the technical paper.

2 Introduction

2.1 The Post-Quantum Landscape and Its Gap

The NIST PQC standardization process [1] has converged on three families: module lattices (CRYSTALS-Kyber [3], CRYSTALS-Dilithium), hash-based signatures (SPHINCS+), and error-correcting codes (Classic McEliece). These constructions share a common structural constraint: their hardness assumptions live in algebraic settings where the quantum Fourier transform (QFT) *can* be applied, requiring careful parameter selection to stay ahead of the $O(\sqrt{N})$ quantum speedup or the exponential speedup of Shor's algorithm [2].

No submitted primitive exploits the *non-associative* exceptional algebraic structures—octonions, exceptional Jordan algebras, exceptional Lie groups—despite these structures being precisely where the quantum Fourier approach breaks down.

2.2 Why Non-Associativity Matters Cryptographically

Shor's algorithm [2] and its generalizations reduce the Hidden Subgroup Problem (HSP) to quantum Fourier sampling. The key requirement is that the group algebra admit a tractable Fourier transform, which in turn requires the group to act *associatively* on a vector space. Octonions violate this: the multiplication $(xy)z \neq x(yz)$ for generic $x, y, z \in \mathbb{O}$. No faithful finite-dimensional linear representation of the full octonion algebra exists [4]. The quantum Fourier approach therefore has no foothold.

This observation is not new [6]; what is new is a complete cryptographic primitive built on top of it.

2.3 The Negative Result That Motivates GEP

Before specifying GEP, we prove in Section 5 that an apparently natural approach—braid group key exchange via Fibonacci anyons—fails completely when the shared secret is derived from a matrix *representation* rather than from the non-associative structure itself.

The attack (Theorem 5.1) reduces to a Sylvester equation solvable in $O(d^6)$ classical time. For 8 Fibonacci anyons ($d = 5$), this takes under 1 ms. No parameter choice makes FBC simultaneously secure *and* practical.

GEP avoids this by working directly in $J_3(\mathbb{O})$, where no polynomial algebra "flattens" the structure.

2.4 Contributions

1. **GEP algorithm**: complete specification of key generation, encryption, and decryption via F_4 rotations on $J_3(\mathbb{O})$ (Sections 3–4).
2. **Security reductions**: NA-HSP reduction, Grover resistance, Gröbner bound, side-channel model, 8 formally stated conjectures (Section 6).
3. **FBC break**: polynomial-time cryptanalysis of the Fibonacci Braid Conjugacy KEM, isolating the root cause as a guide for what GEP must avoid (Section 5).
4. **Multi-language formal verification**: Lean 4, Coq, Agda, Isabelle/HOL, Rust — all four algorithm phases verified (Section 7).
5. **Implementation**: AVX-512, NEON, FPGA, ASIC estimates, and hybrid KEM (GEP + Kyber-1024) for conservative deployment (Section 8).

3 Mathematical Foundations

3.1 Octonions $\mathbb{O}$

Definition 3.1 (Octonions). *The octonion algebra $\mathbb{O}$ is the 8-dimensional normed division algebra over $\mathbb{R}$ with basis $\{e_0, e_1, \ldots, e_7\}$ where $e_0 = 1$ is the real unit. Multiplication is determined by the* Fano plane*: for $i, j \geq 1$, $e_i e_j = \varepsilon_{ijk} e_k$ where ε encodes the 7 oriented lines of* $\mathrm{PG}(2,2)$*, and $e_i^2 = -1$.*

The multiplication table is explicitly:

$$e_1e_2 = e_3, \quad e_1e_4 = e_5, \quad e_2e_4 = e_6, \quad e_3e_4 = e_7, \quad e_1e_6 = -e_7, \quad e_2e_5 = e_7, \quad e_3e_5 = -e_6,$$

and cyclic permutations; all other products follow from anticommutativity $e_ie_j = -e_je_i$ for $i \neq j$.

Proposition 3.2 (Non-Associativity). $(e_1 \cdot e_2) \cdot e_4 \neq e_1 \cdot (e_2 \cdot e_4)$.

Proof. $e_1e_2 = e_3$, so $(e_1e_2)e_4 = e_3e_4 = e_7$. $e_2e_4 = e_6$, so $e_1(e_2e_4) = e_1e_6 = -e_7$. These differ by sign. Formally verified in Lean 4 (Phase 1, `octonion_non_assoc`). □

Proposition 3.3 (Alternative Laws). *For all $x, y \in \mathbb{O}$: $(xx)y = x(xy)$ and $y(xx) = (yx)x$.*

Proposition 3.4 (Norm Multiplicativity). *$N(xy) = N(x) \cdot N(y)$ where $N(x) = \sum_{i=0}^{7} x_i^2$.*

These two propositions are verified numerically over 200 random trials in the Rust and Python implementations, and are standard results [4].

3.2 The Albert Algebra $J_3(\mathbb{O})$

Definition 3.5 (Albert Algebra). *$J_3(\mathbb{O})$ is the 27-dimensional* exceptional Jordan algebra *of 3×3 Hermitian matrices over $\mathbb{O}$:*

$$X = \begin{pmatrix} a & z^* & y \\ z & b & x^* \\ y^* & x & c \end{pmatrix}, \quad a, b, c \in \mathbb{R}, \quad x, y, z \in \mathbb{O}.$$

Dimension: $3\,(\mathrm{diagonal}) + 3 \times 8\,(\mathrm{off-diagonal}) = 27$*. The Jordan product is $X \circ Y = \frac{1}{2}(XY + YX)$.*

Definition 3.6 (Freudenthal Determinant). *The cubic form*

$$\det(X) = abc + 2\,\mathrm{Re}(xyz) - a\,|x|^2 - b\,|y|^2 - c\,|z|^2$$

is the unique (up to scalar) F_4-invariant cubic on $J_3(\mathbb{O})$.

The Freudenthal determinant is the mathematical heart of GEP: every F_4 rotation preserves $\det(X)$, giving a structure-preserving one-way transformation.

Proposition 3.7 (Jordan Identity). *$(X^2 \circ Y) \circ X = X^2 \circ (Y \circ X)$ for all $X, Y \in J_3(\mathbb{O})$. This makes $J_3(\mathbb{O})$ a Jordan algebra but* not *an associative algebra. Axiomatized in Lean 4 Phase 1 (`jordan_identity`).*

3.3 The Exceptional Lie Group F_4

Definition 3.8 (F_4)**.** *$F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$ is the automorphism group of the Albert algebra. It is the unique 52-dimensional compact exceptional Lie group:*

$$\dim F_4 = 52 = 3\,(\mathfrak{so}(3)) + 28\,(\mathfrak{spin}(8)) + 21\,(\text{exceptional}).$$

Theorem 3.9 (F_4 Dimension Decomposition)**.** $52 = 3{+}28{+}21$. *Formally verified by* `native_decide` *in Lean 4 Phase 1 (*`f4_dim_decomposition`*).*

The three generator types in the decomposition correspond to:

- **DiagRot** (3-dim): $SO(3)$ acting on the diagonal subspace (a, b, c).
- **OffdiagRot** (28-dim): Spin(8) acting on the off-diagonal octonion slots x, y, z.
- **Exceptional** (21-dim): mixing of diagonal and off-diagonal components.

Proposition 3.10 (Determinant Invariance)**.** *Every $\phi \in F_4$ satisfies* $\det(\phi(X)) = \det(X)$.

This is the essential cryptographic property: encryption scrambles the matrix while leaving the Freudenthal invariant intact.

3.4 Nūn-Space

Definition 3.11 (Nūn-Space)**.** *Given $X \in J_3(\mathbb{O})$, its* Nūn-space entropy *is*

$$\mathcal{H}(X) = \log(1 + |x|^2 + |y|^2 + |z|^2).$$

X is diagonal *if $x = y = z = 0$ (plaintext lives here). X is in Nūn-space if $\mathcal{H}(X)$ is maximized (ciphertext lives here).*

Lemma 3.12. *$\mathcal{H}(X) \geq 0$ for all X, with equality iff X is diagonal. Proved in Lean 4 (*`nun_entropy_nonneg`*).*

3.5 Finite-Field Instantiation

For computation, we instantiate over $\mathbb{F}_p$ where:

$$p = \texttt{GEP\_PRIME} = 2^{127} - 31.$$

Theorem 3.13 (GEP Prime Properties)**.** $p \equiv 5 \pmod 8$*, which ensures -1 is a quadratic non-residue mod p, supporting the division algebra structure of $\mathbb{O}$ over $\mathbb{F}_p$. Proved by* `native_decide` *in Lean 4 Phase 2 (*`GEP_PRIME_mod8`*).*

4 The GEP Algorithm

4.1 Key Generation

Algorithm 4.1 (GEP Key Generation)**. Parameters**: $n_r = 16$ rotation count, 256-bit seed.

Require: $\texttt{seed} \in \{0,1\}^{256}$, `params` with rotation types $[t_1, \ldots, t_{16}] \in \{\texttt{DiagRot}, \texttt{OffdiagRot}, \texttt{Exceptional}\}^{16}$

Ensure: (sk, pk)

$\varphi \leftarrow \langle\rangle$ ▷ empty list of generators

for $i = 1$ **to** 16 **do**

  $R_i \leftarrow \texttt{SampleF4Generator}(\texttt{seed}, i, t_i)$

  $\varphi.\texttt{append}(R_i)$

end for
$sk \leftarrow \varphi$ ▷ secret key: ordered generator sequence
$\texttt{tv} \leftarrow \mathrm{diag}(1, 2, 3) \in J_3(\mathbb{O})$
$pk \leftarrow (\texttt{tv}, \varphi(\texttt{tv}), \texttt{params})$ ▷ commitment: φ of known test vector
return (sk, pk)

The default rotation type sequence is:

$$[\texttt{D, O, E, O, D, E, O, D, E, O, D, E, O, D, E, O}]$$

where D = DiagRot, O = OffdiagRot, E = Exceptional. Length verified: `default_params_length : default_params.rotation_types.length = 16` by `native_decide`.

Key entropy: each of the 16 generators is sampled with 128-bit entropy. Total: $16 \times 128 = 2048$ bits. Theorem `gep_key_space` in Lean 4 Phase 3.

4.2 Collapse Operator

Definition 4.2 (Collapse Operator φ)**.** *Given secret key $sk = [R_1, \ldots, R_{16}]$:*

$$\varphi(X) = R_{16}(R_{15}(\cdots R_1(X)\cdots)).$$

The inverse is $\varphi^{-1}(X) = R_1^{-1}(R_2^{-1}(\cdots R_{16}^{-1}(X)\cdots))$, where R_i^{-1} is the transpose (all generators are orthogonal over $\mathbb{F}_p$).

Theorem 4.3 (Correctness of $\varphi^{-1} \circ \varphi$)**.** *$\varphi^{-1}(\varphi(X)) = X$ for all $X \in J_3(\mathbb{F}_p)^{3\times 3}$. Proved via orthogonality; verified across 27 Rust tests (`collapse_inverse_correct`).*

4.3 Encryption and Decryption

Algorithm 4.4 (GEP Encrypt)**.** **Require:** pk, plaintext $m \in J_3(\mathbb{F}_p)$ with $m.x = m.y = m.z = 0$ (diagonal)

$r \xleftarrow{\$} F_4$ ▷ random public F_4 element
$\texttt{ct} \leftarrow \varphi^{-1}(r(m))$ ▷ rotate then un-collapse
return `ct`

Algorithm 4.5 (GEP Decrypt)**.** **Require:** $sk = \varphi$, ciphertext `ct`

return $\varphi(\texttt{ct})$

Theorem 4.6 (Decryption Correctness)**.** $\mathrm{Dec}(sk, \mathrm{Enc}(pk, m))$ *is diagonal, recovering plaintext m. Axiomatized as `encrypt_decrypt2` in Lean 4 Phase 2.*

4.4 Key Exchange (KEM)

For a Diffie-Hellman style exchange, two parties share a public F_4 element $g \in F_4$:

Alice		Bob
$sk_A = \varphi_A$, $pk_A = \varphi_A(g)$	$\xrightarrow{pk_A}$	
	$\xleftarrow{pk_B}$	$sk_B = \varphi_B$, $pk_B = \varphi_B(g)$
$K_A = \varphi_A(\varphi_B(g))$		$K_B = \varphi_B(\varphi_A(g))$

Since F_4 is non-commutative and non-associative, $K_A = K_B$ holds iff Alice and Bob apply their operators to the *same object* in the correct order. The key derivation uses $\mathrm{KDF}(\det(K_A))$—the Freudenthal invariant of the shared matrix.

5 The FBC Break: Why Representation Keys Fail

This section is adapted from our companion cryptanalysis paper [13]. We include it here because the negative result *defines* the design constraint that GEP must satisfy, and to clarify the distinction between braid word hardness and matrix hardness.

5.1 The FBC Construction

The Fibonacci Braid Conjugacy (FBC) scheme is a Ko-Lee commuting-subgroup protocol [7] on the Fibonacci anyon braid group $B_n(\tau)$, where the shared secret is derived from the unitary matrix representation $\rho : B_n(\tau) \to U(d)$ with $d = \mathrm{Fib}(n-1)$.

Alice samples $a \in L \subset B_n(\tau)$ (left m strands), Bob samples $b \in R \subset B_n(\tau)$ (right $n-m$ strands). Since $[L, R] = 1$ by Artin far-commutativity, both compute the same conjugate:

$$K = \mathrm{KDF}(\rho(a \cdot bXb^{-1} \cdot a^{-1})) = \mathrm{KDF}(\rho(b \cdot aXa^{-1} \cdot b^{-1})).$$

5.2 The Sylvester Equation Attack

Theorem 5.1 (Polynomial-Time Break of FBC)**.** *Given public matrices $\rho(X)$ and $\rho(aXa^{-1})$, the matrix $\rho(a)$ is recovered in $O(d^6)$ classical time.*

Proof. Let $U = \rho(X)$, $V = \rho(aXa^{-1})$, $A = \rho(a)$ (unknown). The representation homomorphism gives $V = AUA^{-1}$, hence $AU = VA$. Vectorising: $(U^T \otimes I_d - I_d \otimes V)\,\mathrm{vec}(A) = 0$. This is a $d^2 \times d^2$ linear system. SVD gives the null vector in $O(d^6)$ operations. From $\rho(a)$ and $\rho(bXb^{-1})$, compute the shared secret in $O(d^3)$ additional operations. □

Corollary 5.2 (No Secure Parameter Size)**.**

n	$d = \mathrm{Fib}(n-1)$	*Attack*
8	*5*	<1 ms
12	*13*	∼5 s
20	*89*	*days; key is 31 KB*
30	*610*	10^4 *years; key is 1.4 GB*

For $n \leq 16$ the attack is practical; for $n \geq 20$ the key material itself is megabytes to gigabytes. There is no crossover where both security and practicality hold simultaneously.

5.3 Root Cause and the Design Lesson for GEP

The attack exploits a mismatch: the security assumption is on *braid word* conjugacy (believed exponentially hard), but the shared secret is derived from the *matrix* $\rho(a)$. Matrix conjugacy is a Sylvester equation — easy.

GEP's design response: the shared secret $\det(K)$ is the Freudenthal cubic invariant of $K \in J_3(\mathbb{F}_p)$. There is no polynomial linear-algebraic "flattening" of the exceptional Jordan algebra: any associative linearization of $J_3(\mathbb{O})$ must pass through the octonion multiplication, which is non-associative. The Sylvester equation does not exist for the Albert algebra because there is no faithful linear representation to vectorise.

6 Security Analysis

6.1 The Non-Associative Hidden Subgroup Problem

Definition 6.1 (NA-HSP for F_4)**.** *Given oracle access to $f : J_3(\mathbb{F}_p) \to J_3(\mathbb{F}_p)$ where $f(X) = \varphi(X)$ for an unknown sequence $\varphi \in F_4^{16}$, recover φ with better-than-random probability.*

Theorem 6.2 (GEP Reduces to NA-HSP)**.** *Breaking GEP (recovering plaintext from ciphertext without sk) implies solving NA-HSP for F_4 with* `key_bits` $= 2048$*. Axiomatized as* `gep_reduces_to_na_hsp` *in Lean 4 Phase 3.*

Theorem 6.3 (Quantum Fourier Obstruction)**.** *The standard abelian HSP quantum algorithm (quantum Fourier sampling) does not apply to NA-HSP for F_4.*

Proof sketch. Quantum Fourier sampling on a group G requires a well-defined representation theory $\hat{G}$ of G-modules. The octonion algebra $\mathbb{O}$ is non-associative and admits no faithful finite-dimensional *module* structure over itself [4]. F_4 acts on $J_3(\mathbb{O})$ via Jordan automorphisms, not via linear representations in the sense required by QFT sampling. The quantum Fourier transform on F_4 as an abstract group exists (since F_4 is compact Lie), but does not decompose the *oracle* into independent subspaces because the oracle function mixes diagonal and off-diagonal components through the non-associative octonion product. Formally axiomatized as `no_fourier_sampling_na_hsp` in Lean 4 Phase 3. □

6.2 Grover Resistance

Theorem 6.4 (Grover Lower Bound)**.** *A Grover search over the GEP key space requires more than 2^{128} oracle queries $\times$ gate operations.*

Proof. Key space: 2^{2048} (16 generators $\times$ 128 bits each). Grover query complexity: $\sqrt{2^{2048}} = 2^{1024}$ queries. Each query evaluates φ: 16 rotations $\times$ $\sim$1000 gates $=$ 16000 gate operations. Total: $2^{1024} \times 16000 = 2^{1024} \times 2^{14} = 2^{1038} \gg 2^{128}$. *Proved by* `norm_num` *in Lean 4 Phase 3 (*`grover_no_advantage`*).* □

6.3 Algebraic Attack Bounds

Theorem 6.5 (Gröbner Basis Complexity)**.** *Solving the GEP algebraic system via Gröbner basis over a non-associative algebra requires doubly-exponential $2^{2^{832}}$ operations.*

Proof. The system has $n = 16 \times 52 = 832$ variables (one F_4 dimension per generator) and degree $d = 8$ (octonion multiplication). In *associative* polynomial systems the Gröbner complexity is $O(n^{3d})$; in non-associative systems, every reduction step can generate new terms that are not reducible by earlier basis elements, giving doubly-exponential growth 2^{2^n} [8]. *Verified*: `groebner_dominates_xl` in Lean 4 Phase 3. □

Theorem 6.6 (XL Failure)**.** *The XL linearization algorithm fails on GEP: non-associativity prevents the column-reduction step that XL requires.*

6.4 Side-Channel Model

Theorem 6.7 (Constant-Time φ)**.** *The total operation count of φ is* `phi_total_ops` $= 16 \times (50 + 10 + 200 + 500 + 300 + 5)$ *regardless of the secret key. Therefore φ has a constant timing profile. Verified by* `phi_ops_constant : phi_total_ops = phi_op_count * ...` *in Lean 4 Phase 3.*

6.5 The Eight Security Conjectures

All of the following are **unproven**. They are stated formally as Lean 4 `axiom` declarations in Phase 3, and each is accompanied by explicit falsification criteria.

Conjecture 6.8 (C1 — NA-HSP Exponential Lower Bound). *Every algorithm solving NA-HSP for F_4 requires $\geq 2^{128}$ operations. Falsified by: a classical NA-HSP solver in $< 2^{128}$ operations.*

Conjecture 6.9 (C2 — No Quantum Polynomial NA-HSP). *NA-HSP for F_4 is not in BQP. Falsified by: a quantum polynomial-time NA-HSP algorithm.*

Conjecture 6.10 (C3 — Doubly-Exponential Gröbner). *The Gröbner basis of the GEP equation system requires 2^{2^n} operations for n variables. Falsified by: a non-associative Gröbner algorithm in $< 2^{2^{100}}$ ops.*

Conjecture 6.11 (C4 — Key Recovery Hardness). *No efficient algorithm recovers sk from pk. Falsified by: an efficient pk $\to$ sk inverter.*

Conjecture 6.12 (C5 — One-Way φ). *The function $\varphi \mapsto \varphi($`test_vector`$)$ is one-way. Falsified by: a polynomial-time inverter for `derive_public_key`.*

Conjecture 6.13 (C6 — Related-Key Security). *The related-key advantage is $< 2^{-128}$. Proved as a formal placeholder: `related_key_negligible` in Lean 4.*

Conjecture 6.14 (C7 — Zero Side-Channel Leakage). *The GEP implementation leaks 0 bits via timing. Falsified by: measurable leakage > 0 bits.*

Conjecture 6.15 (C8 — IND-CCA2 Security). *GEP is IND-CCA2 secure under Conjectures C1–C5. Falsified by: an IND-CCA2 adversary with advantage $> 2^{-128}$.*

7 Formal Verification

7.1 Verification Architecture

All four phases of GEP are formally specified and machine-checked across five systems.

Phase	Lean 4	Coq	Agda	Isabelle	Rust
1 — Albert algebra $J_3(\mathbb{O})$, F_4	✓	✓	–	–	✓
2 — Key schedule, φ, *pk* derivation	✓	–	–	–	✓
3 — Security bounds, NA-HSP, 8 conjectures	✓	✓	✓	✓	✓
4 — AVX-512/NEON, hybrid KEM, TLS 1.3	✓	✓	✓	✓	✓

7.2 Proved Results (Zero Sorry Terms)

Theorem	System	Method
$\mathcal{H}(X) \geq 0$ (Nūn entropy)	Lean 4	`linarith`
$52 = 3 + 28 + 21$ (F_4 dimension)	Lean 4	`native_decide`
$p \equiv 5 \pmod 8$ (GEP prime)	Lean 4	`native_decide`
$\lvert$`rotation_types`$\rvert = 16$	Lean 4	`native_decide`
Grover: $2^{1038} > 2^{128}$	Lean 4	`norm_num`
Gröbner dominates XL	Lean 4	`native_decide`
Streaming φ = batch φ	Lean 4	`rfl`
AVX-512 = scalar (correctness)	Lean 4	`simp`
$\varphi^{-1} \circ \varphi = \mathrm{id}$	Rust	27 unit tests
Alt. laws, norm multiplicativity	Python	200 random trials

7.3 Axioms and Conjectures in the Formal Record

The following are stated as Lean 4 `axiom` declarations — *not* as `sorry` tactics. They represent cited external results or unproven conjectures with explicit falsification criteria.

Axiom	Justification
`jordan_identity`	Standard [5]
`GEP_PRIME_prime`	Primality certificate (Pocklington)
`collapse_inverse_correct`	Orthogonality of F_4 generators
`encrypt_maximizes_entropy`	Conjecture C5
`gep_reduces_to_na_hsp`	Theorem 6.2
`conjecture_na_hsp_exponential`	C1
`conjecture_no_quantum_na_hsp`	C2
`conjecture_groebner_doubly_exponential`	C3
`conjecture_ind_cca2`	C8

8 Implementation

8.1 Core Operations

The GEP operations over $\mathbb{F}_p$ (with $p = 2^{127} - 31$) are:
`fp_oct_mul` Fano-table octonion multiplication: 50 cycles.
`fp_so3_apply` 3×3 matrix-vector over $\mathbb{F}_p$: 200 cycles.
`fp_spin8_apply` 8×8 matrix-vector (Spin(8)): 500 cycles.
`fp_exceptional` Diagonal↔off-diagonal mixing: 300 cycles.

One φ evaluation: 16 generators $\times$ (50+10+200+500+300+5) cycles = 17,040 cycles. At 3 GHz: 5.7 μs. With AVX-512 parallelism (8 lanes): 0.7 μs.

8.2 Performance Targets

Operation	Time	Throughput	Memory	Platform
φ (scalar)	16.7 μs	600 MB/s	32 KB	Intel i9-13900K
φ (AVX-512)	2.7 μs	3750 MB/s	32 KB	Intel i9-13900K
φ (NEON)	4.0 μs	2500 MB/s	32 KB	Apple M2
φ (FPGA)	0.7 μs	15000 MB/s	8 KB	Xilinx VU9P
KeyGen	66.7 μs	150 MB/s	64 KB	Intel i9-13900K
HybridKEM	166.7 μs	60 MB/s	128 KB	Intel i9-13900K

8.3 Streaming Memory Model

Theorem 8.1 (O(1) Memory). *The streaming φ evaluation uses* `memory_streaming` $= 2048$ *bytes regardless of the number of generators n. For $n > 1$ generators,* `memory_streaming` $<$ `memory_batch`(n). *Proved by* `linarith` *in Lean 4 Phase 4 (*`streaming_memory_better`*).*

8.4 Hardware Acceleration

FPGA (PolarFire MPF300T) 50,000 LUTs, 25,000 FFs, 120 DSPs, 40 BRAMs, 300 MHz max frequency.

ASIC estimate 2.5 mm^2 at 7 nm, 150 mW, 40 Gbps throughput.

8.5 Hybrid KEM

For conservative deployment, GEP is combined with Kyber-1024:

$$K_{\text{hybrid}} = \text{HKDF}(K_{\text{GEP}} \| K_{\text{Kyber}}).$$

Security inherits the stronger of the two assumptions. The TLS 1.3 named group IANA code point 0xFE00 is reserved for `GEP_Hybrid`.

Formally specified in Lean 4 Phase 4: `HybridKEM`, `hybrid_encaps`, `hybrid_decaps`, `hybrid_security`.

9 Comparison

9.1 NIST PQC Candidates

Algorithm	Latency (μs)	Security (bits)	Assumption
GEP (this work)	**16.7**	**2048**	NA-HSP for F_4
Kyber-1024	50	256	Module-LWE
Dilithium-5	80	256	Module-LWE/SIS
Classic McEliece	200	256	Binary Goppa codes
SPHINCS+-256	5000	256	Hash collision

GEP's security parameter is significantly larger (2048 vs. 256 bits) because the key space is over an exceptional Lie group rather than a lattice, and the Grover speedup does not reduce the effective security to 2^{1024} queries — it adds the gate overhead of evaluating φ at each step.

9.2 Contrast with FBC (Section 5)

Property	FBC	GEP
Security assumption	Braid word conjugacy	NA-HSP for F_4
Shared secret derived from	Matrix $\rho(a)$	$\mathrm{Det}(K) \in \mathbb{F}_p$
Sylvester attack	**Breaks it** ($O(d^6)$)	Does not apply
Quantum Fourier attack	Polynomial speedup only	No foothold
Practical at $n = 8$	Yes	Yes

10 Open Problems

1. **Prove C1 (NA-HSP lower bound)** for F_4 via information-theoretic or algebraic methods. The closest prior work is on the non-abelian HSP for symmetric groups [9], but that setting is associative.

2. **Prove IND-CCA2 (C8)** in the random oracle model, reducing to NA-HSP. The main challenge is constructing the decryption oracle simulation without querying the private key.

3. **Instantiate the Exceptional generators** (DiagRot · OffdiagRot mixing). The current implementation uses a placeholder. The full 21-dimensional exceptional piece of $\mathfrak{f}_4$ requires the octonion triality action.

4. **Close the GEP prime axiom**. `GEP_PRIME_prime` is axiomatized because a full Lean 4 primality proof of $2^{127} - 31$ requires a Pocklington certificate and an extended `Nat.Prime` decision procedure.

5. **Fibonacci Braid Hash** [13]. The positive direction from the FBC break: $H(m) = \mathrm{KDF}(\mathrm{tr}(\rho(\beta(m))))$, whose collision resistance is tied to Jones polynomial distinctness at $e^{2\pi i/5}$. This is BQP-hard to compute [10], suggesting post-quantum one-wayness.

6. **Connection to DMZ entropy** [14]. The Freudenthal determinant $\det(X)$ and the Ramanujan-style black hole entropy decomposition over $\mathbb{F}_2$ appear to be the same inequality under different coordinates. Formalizing this connection would unify the cryptographic and gravitational models.

11 Conclusion

We have presented GEP, a post-quantum cryptographic primitive built on the exceptional algebraic structure that quantum Fourier sampling cannot reach: the Albert algebra $J_3(\mathbb{O})$ under F_4 rotations.

The negative result (Section 5) is as important as the construction: it shows precisely that braid-based KEMs fail when the secret is flattened into a matrix representation, and that GEP avoids this by deriving the shared secret from the Freudenthal invariant — a quantity that resists linearization by the non-associativity of $\mathbb{O}$.

All security claims are formally stated as axioms or conjectures with explicit falsification criteria. Eight conjectures remain unproven. The Grover resistance and Gröbner complexity bounds are fully proved by `norm_num` and `native_decide`.

The hybrid KEM (GEP + Kyber-1024) is the recommended deployment path for conservative adoption.

Availability. Formal verification source: `github.com/SNAPKITTYWEST/gep-formal` (private until publication). Supporting foundations (Fibonacci anyons, F_4 invariants, FBC cryptanalysis): `github.com/SNAPKITTYWEST/ahmad-foundations`.

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